

HYPOTHESIS TESTING (PART 2)

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MORE HYPOTHESES? IS THAT RIGHT?

In Article No.20, we established the core fundamentals behind hypothesis testing, therefore, we can continue to explore how these concepts are applied in practice. To analyse real world data, we often ask different types of questions, such as: Are two variables related? Do two separate groups differ from one another? Does a treatment produce a change in the same subjects over time? Let us introduce several more hypothesis tests designed to answer these types of questions, focusing on tests of independence, comparisons of independent groups, and paired sample tests.

TESTING FOR INDEPENDENCE

How do we test whether two jointly distributed random variables are independent of each other. The idea of this test is similar to that of the test of goodness of fit; essentially, we check if the assumption of independence provides a good fit. Suppose that n IID pairs with the same distribution as discrete random variables (X, Y) are observed. Suppose that the frequencies of the different values of the pair are tabulated as:

$Y \backslash X$	x_1	x_2	$\dots$	x_m	Total
y_1	f_{11}	f_{21}		f_{m1}	$f_{.1}$
y_1	f_{12}	f_{22}		f_{m2}	$f_{.2}$
$\vdots$			$\ddots$		$\vdots$
y_n	f_{1n}	f_{2n}		f_{mn}	$f_{.n}$
Total	$f_{1.}$	$f_{2.}$	$\dots$	$f_{m.}$	n

From above, we can see that the estimates of the probabilities, $P(X = x_i, Y = y_j)$, would be $\frac{f_{ij}}{n}$, and the estimates of the marginal probabilities, $P(X = x_i)$ and $P(Y = y_j)$ would be $\frac{f_{i.}}{n}$ and $\frac{f_{.j}}{n}$, respectively. However, if X and Y are independent, then $P(X = x_i, Y = y_j) = P(X = x_i) \times P(Y = y_j)$, thus, we expect $(i, j)^{\text{th}}$ cell to be around $f_{ij} \approx \frac{f_{i.} \times f_{.j}}{n}$. Therefore, to test $H_0 : X$ and Y are independent against $H_1 : X$ and Y are not

independent, we can use the Pearson's χ^2 statistic, that is $\chi_0^2 = \sum_{\text{all cells}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \sum_{\text{all cells}} \frac{\left(f_{ij} - \frac{f_{i.} \times f_{.j}}{n}\right)^2}{\left(\frac{f_{i.} \times f_{.j}}{n}\right)}$, and reject for large values of χ_0^2 . Note that here we effectively fit $(m-1) + (n-1)$ marginal probabilities. Thus, we obtain $d.f. = mn - ((m-1) + (n-1)) - 1 = (m-1)(n-1)$.

COMPARING TWO INDEPENDENT SAMPLES

Let us compare two samples from normal distributions. To facilitate calculations, we would assume that the samples have the same variance, but possibly different expectations. How can we check if the two samples are drawn from populations with the same mean, which would mean that they are from the same distribution? Consider n IID observations $X_1, X_2, \dots, X_n$ from $N(\mu_1, \sigma^2)$ and m IID observations $Y_1, Y_2, \dots, Y_m$ from $N(\mu_2, \sigma^2)$, where μ_1 and μ_2 unknown. Set $H_0 : \mu_1 = \mu_2$, and $H_1 : \mu_1 \neq \mu_2$.

Therefore, $\bar{X} \sim N\left(\mu_1, \frac{\sigma^2}{n}\right)$ and $\bar{Y} \sim N\left(\mu_2, \frac{\sigma^2}{m}\right)$. Since the two samples are independent, $\bar{X}$ and $\bar{Y}$ are independent, giving $\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \sigma^2\left(\frac{1}{n} + \frac{1}{m}\right)\right)$.

If σ^2 is known, then a C.I. for $\mu_1 - \mu_2$ could be based on $Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n} + \frac{1}{m}}}$, which follows a standard normal distribution. Thus, a $100(1 - \alpha)\%$ C.I. for $\mu_1 - \mu_2$ would be $(\bar{X} - \bar{Y}) \pm z_{\frac{\alpha}{2}} \sigma \sqrt{\frac{1}{n} + \frac{1}{m}}$. We accept H_0 if $0 \in (\bar{X} - \bar{Y}) \pm z_{\frac{\alpha}{2}} \sigma \sqrt{\frac{1}{n} + \frac{1}{m}}$, and reject otherwise.

If σ^2 is unknown, which is generally the case, then we first estimate by $s_p^2 = \frac{(n-1)s_X^2 + (m-1)s_Y^2}{n+m-2}$, where $s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ and $s_Y^2 = \frac{1}{m-1} \sum_{i=1}^m (Y_i - \bar{Y})^2$. Note that $\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2$ and $\frac{\sum_{i=1}^m (Y_i - \bar{Y})^2}{\sigma^2} \sim \chi_{m-1}^2$. Since they are independent, we have $\frac{(\sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^m (Y_i - \bar{Y})^2)}{\sigma^2} \sim \chi_{n+m-2}^2$. Also, we have that $\sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^m (Y_i - \bar{Y})^2$ is independent of $\bar{X} - \bar{Y}$, and thus of Z , which means the statistic $t = \frac{Z}{s_p} = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n} + \frac{1}{m}}} / \frac{s_p}{\sigma} =$

$\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{s_p\sqrt{\frac{1}{n}+\frac{1}{m}}}$ follows a t distribution with $(n+m-2)$ degrees of freedom. Thus, a $100(1-\alpha)\%$ C.I. for $\mu_1 - \mu_2$ would be $(\bar{X} - \bar{Y}) \pm t_{n+m-2; \frac{\alpha}{2}} s_p \sqrt{\frac{1}{n} + \frac{1}{m}}$. We accept H_0 if $0 \in (\bar{X} - \bar{Y}) \pm t_{n+m-2; \frac{\alpha}{2}} s_p \sqrt{\frac{1}{n} + \frac{1}{m}}$, and reject otherwise.

COMPARING PAIRED SAMPLES

In many experiments, the samples are paired, for example, consider the runs scored by a group of English batsmen in two innings of a test match. Therefore, we often want to complete a hypothesis test based on comparing these paired samples. Consider n IID pairs $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$, allowing the possibility that X_i and Y_i can be dependent for same i , but independent whenever $i \neq j$. We want to check if $\mu_1 = \mu_2$, where $\mathbb{E}[X_i] = \mu_1, \mathbb{E}[Y_i] = \mu_2 \forall i$ and j . Let $D_i = X_i - Y_i$, and test $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$. Note that $\mathbb{E}[D_i] = \mu_1 - \mu_2 = \mu_D$ and $Var(D_i) = Var(X_i - Y_i) = Var(X_i) + Var(Y_i) - 2Cov(X_i, Y_i) = \sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY} = \sigma_D^2$. Let us assume that the D_i 's are normal, that is, $\bar{D} \sim N(\mu_1 - \mu_2, \frac{1}{n}(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY})) = N(\mu_D, \frac{\sigma_D^2}{n})$. Thus, a two-sided C.I. for μ_D is $\left[\bar{D} \pm z_{\frac{\alpha}{2}} \left(\frac{\sigma_D}{\sqrt{n}} \right) \right]$. If σ^2 's are unknown, we must estimate via sample variances.

CONCLUSION

Hypothesis testing is versatile, it can be applied to a range of questions. The tests discussed in this article and the previous illustrate how hypothesis testing can be adapted to answer these questions. Choosing the correct test ensures that conclusions are both meaningful and statistically sound.

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